

First Writers-Readers Problem in a Microcontroller

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Selected problem:

Writer-reader problem (Version 1). Defined by Silberschatz, et.al as a variation of the readers-writers problem that requires that no reader be kept waiting unless a writer has already obtained permission to use the shared object. Therefore, it is a version that favors the reader.

This will be implemented in a ESP32-WVROOM-32D microcontroller, The main goal is to isolate the reading of different sensors along with the handling of their values and the writing of these values to an SD or MicroSD card. The ultimate goal of this program is to read from the sensors as fast as possible and log these values on the SD card to be handled later. The complete build will be part of a rocket payload and it will help to determine possible survivability metrics and the position of the rocket.

Intended Approach to Solving the Problem:

Research will be done by understanding lecture notes, the book material and website resources. We will use an available thread library such as POSIX Threads pthreads and the esp_thread.h library as it is supported by Espressif-IDF. Additionally we will initially use two MPU6050 Accelerometers as main sensors with the possibility of adding more. Writing to the SD card will be handled with the SDMMC Host API or the SPI Host API and other SD association provided freeware. Reading the accelerometers and other sensors is to be done primarily with I2C communication. The different turnaround times of different tasks will be analyzed to correctly optimize the system.

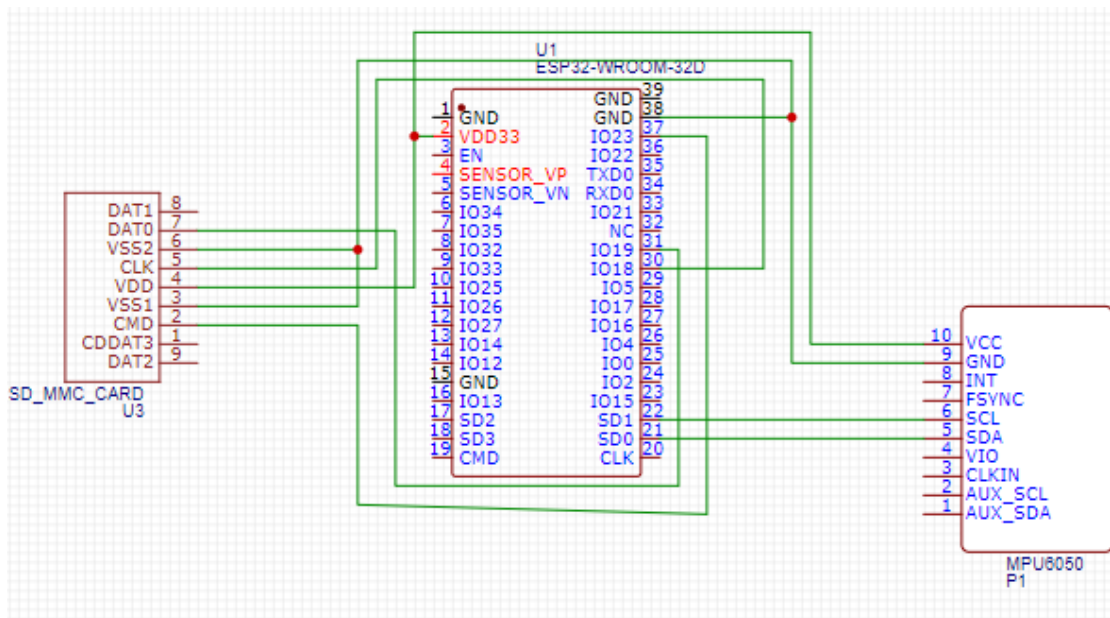


Figure 1: Minimized simple system.

Methods for Evaluating Results:

Statistical tests will be used to determine if any starvation is occurring and if there is any statistically significant difference from the datasheet effectiveness expected, this can help confirm whether there is a correct implementation of the problem. The system must also be sufficiently redundant and safe to work in a rocket payload for an extended amount of time.